

PUBLISHED: 27-JUN-2014
2014.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - V8 S/C 5.0L PETROL

LEFT FUEL RAIL (G1509466)

REMOVAL AND INSTALLATION

19.60.03	FUEL RAIL - LH BANK - RENEW	5000 CC, AJ V8, SUPERCHARGED	1.3	USED WITHINS	
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SPECIAL TOOL(S)



E114526

310-197

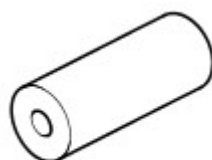
Remover, Fuel
Injector



E107680

310-198

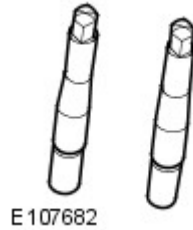
Installer,
Teflon Seal



E107681

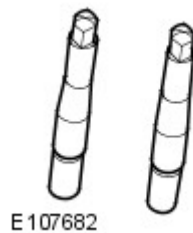
310-199

Re-shape Tool, Teflon
Seal



310-200-01

Fuel Rail Installation Guide
Pins - Threaded



310-200-02

Fuel Rail Installation Guide
Pins - Unthreaded

REMOVAL



CAUTIONS:

- Make sure that tools and equipment are clean and free of foreign material and lubricant.
- Always carry out the cleaning process before carrying out any repairs to the fuel injection system components. Failure to follow this instruction may result in foreign matter ingress to the fuel injection system.



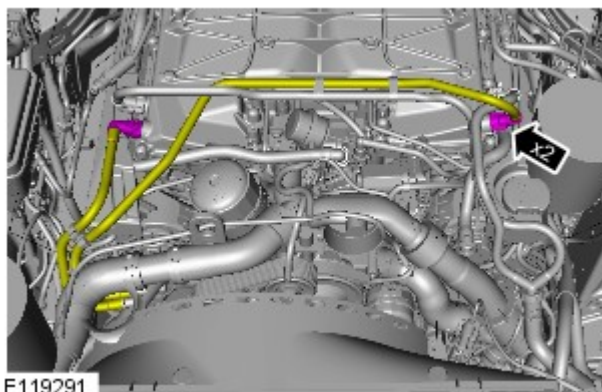
NOTE:

Some variation in the illustrations may occur, but the essential information is always correct.

1. Refer to: Petrol and Petrol-Ethanol Fuel Systems Health and Safety Precautions (100-00 General Information, Description and Operation).
2. Refer to: Fuel System Pressure Release - V8 5.0L Petrol/V8 S/C 5.0L Petrol (310-00 Fuel System - General Information, General Procedures).
3. Disconnect the battery ground cable.
Refer to: Specifications (414-00, Specifications).
4. Refer to: Fuel Injection Component Cleaning (303-04 Fuel Charging and Controls - V8 5.0L Petrol, General Procedures).
5. Refer to: Air Cleaner Outlet Pipe T-Connector (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation).

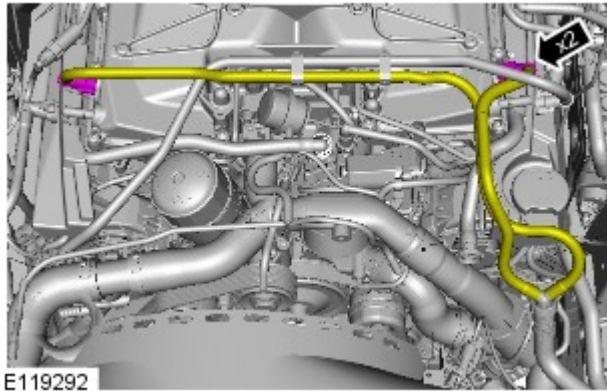
6. **⚠ CAUTION:**

Be prepared to collect escaping fluids.



7. **⚠ CAUTION:**

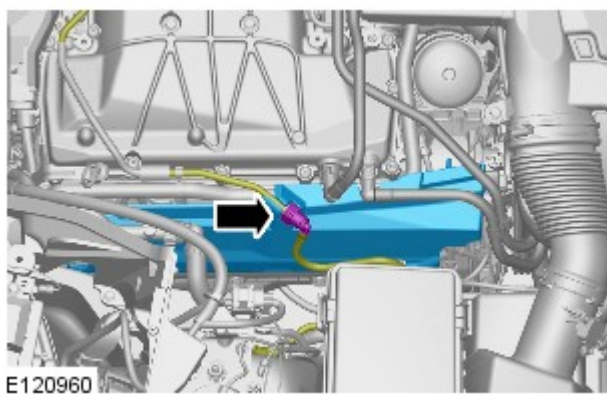
Be prepared to collect escaping fluids.



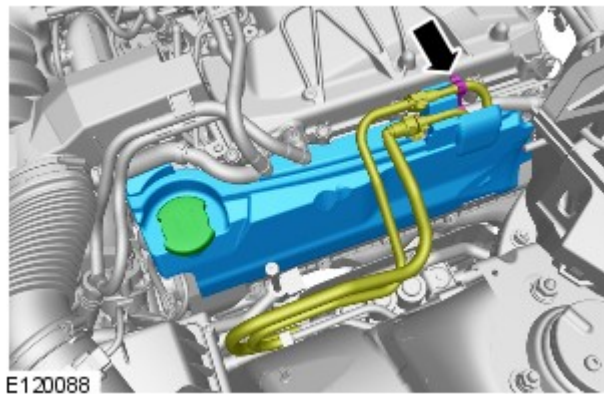
8.

⚠ CAUTION:

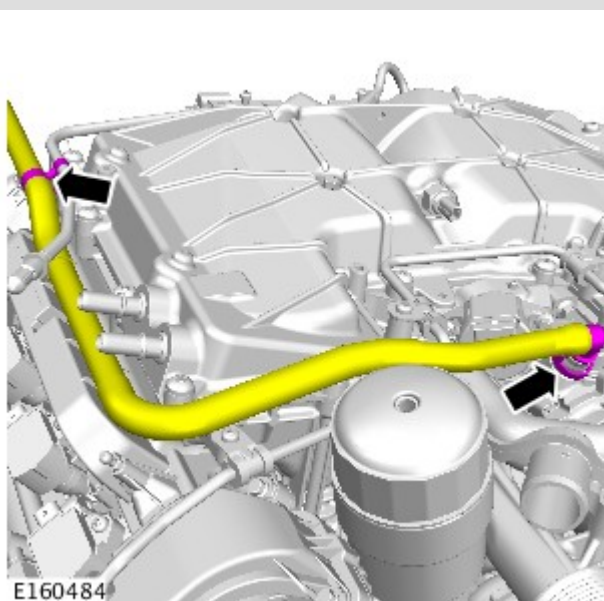
Be prepared to collect escaping fluids.



9.



10.



11.



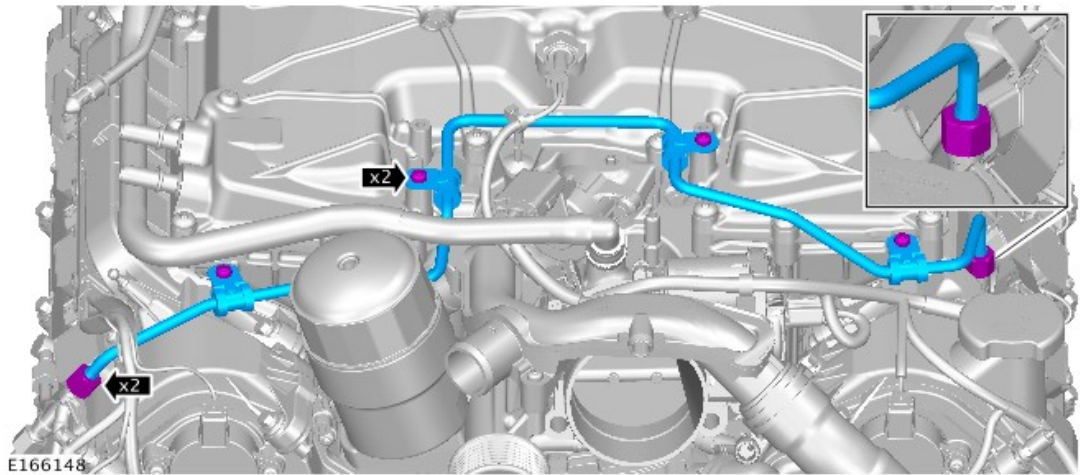
WARNING:

Do not smoke or carry lighted tobacco or open flame of any type when working on or near any fuel related components. Highly flammable vapors are always present and may ignite. Failure to follow these instructions may result in personal injury.

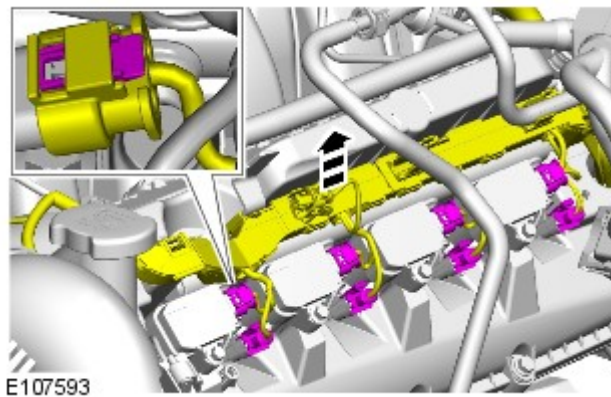


CAUTIONS:

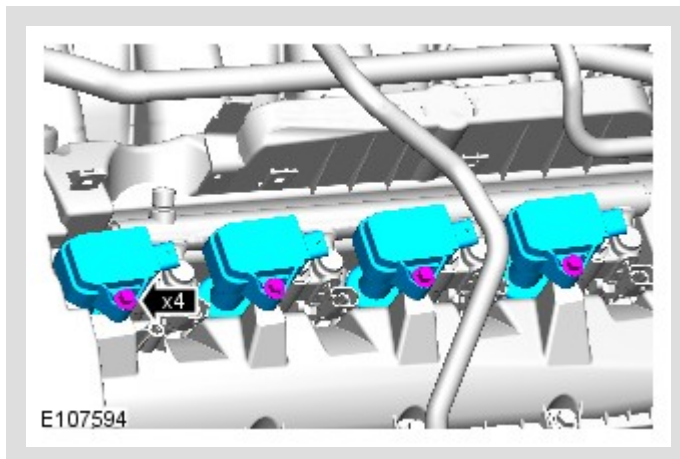
- Make sure that the fuel line union does not rotate.
- Make sure that all openings are sealed. Use new blanking caps.
- Be prepared to collect escaping fluids.



12.



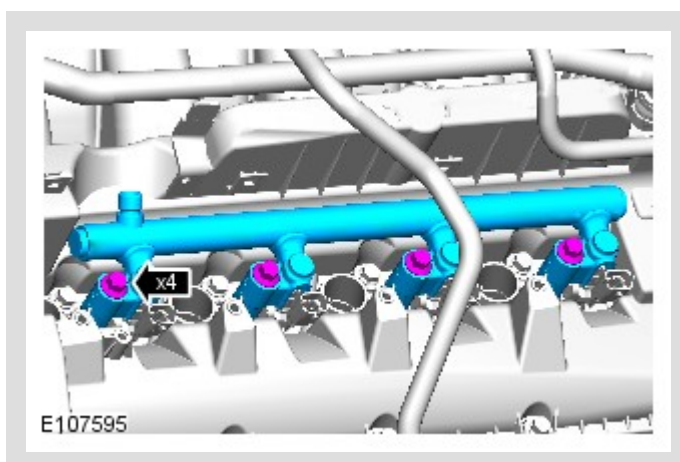
13.



14.

⚠ CAUTION:

Make a note of the fuel injector clamp alignment to the fuel rail prior to removal.



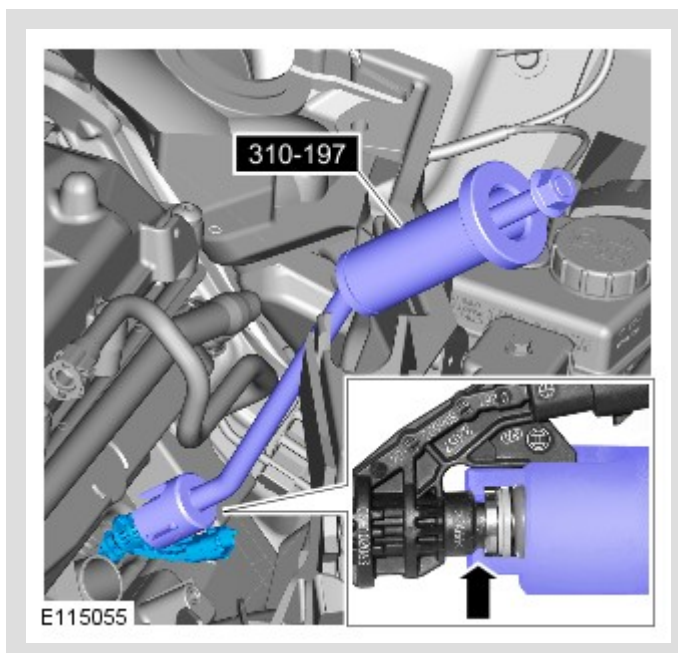
15.

⚠ CAUTIONS:

- Make sure that the special tool is located correctly to the fuel injector prior to removing the fuel injector.
- Make sure that the special tool is held square to the fuel injector during removal.
- Make sure that all open ports are covered to prevent any foreign material ingress.

 NOTE:

RH illustration shown, LH is similar.

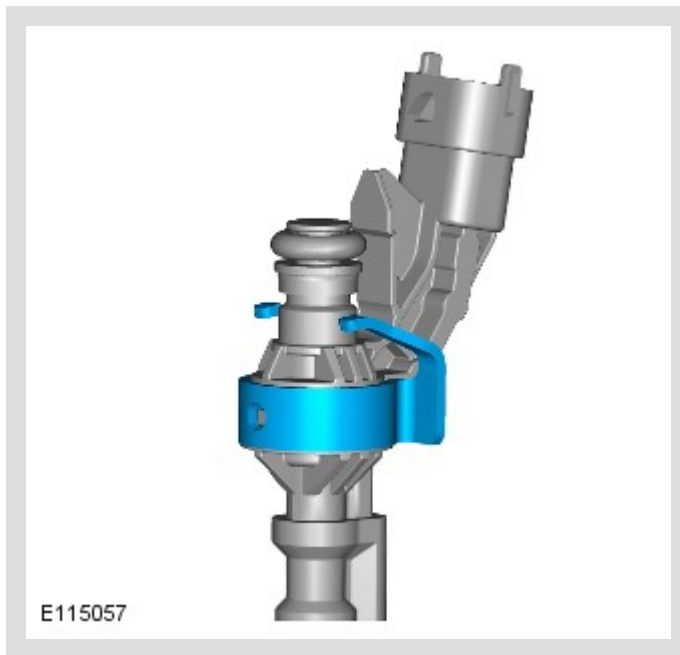


Special Tool(s): [310-197](#)

16.

 CAUTIONS:

- If new fuel injectors are installed, a new injector clamp must be installed.
- If the fuel injector is being removed without a new component being installed, the fuel injector clamp must remain with the fuel injector it is removed with.

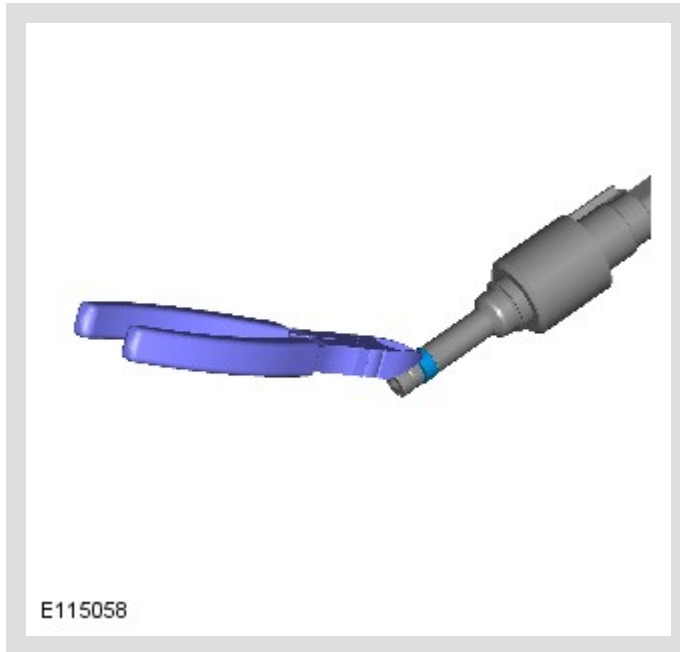


17.



CAUTIONS:

- Do not use a knife to remove the Teflon seal as damage could occur to the fuel injector.
- Do not cut the Teflon seal too deep as damage could occur to the fuel injector.
- Pinch the Teflon seal to allow the tool to cut the Teflon seal without damaging the fuel injector.

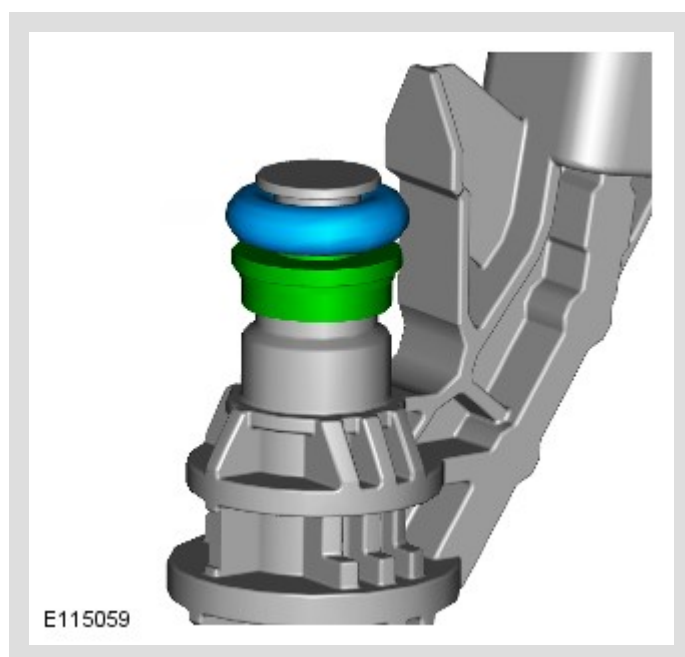


18.



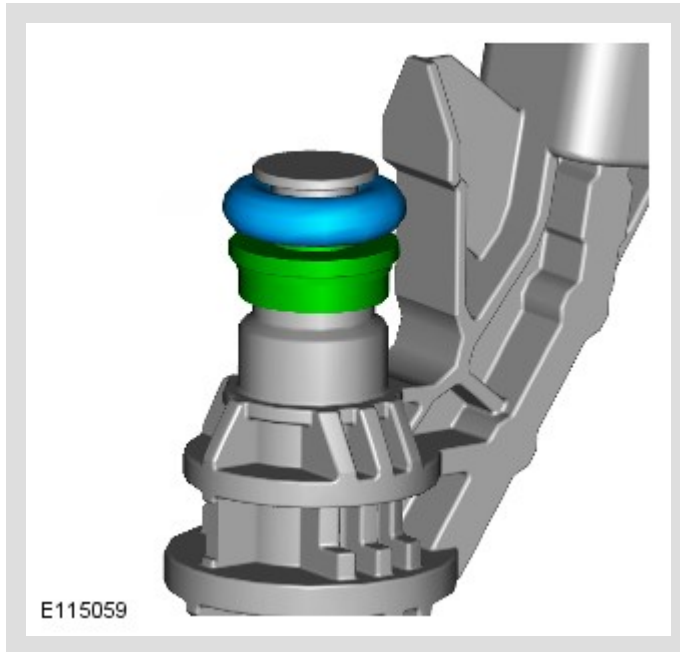
CAUTION:

Do not use any sharp tools to remove the O-ring seal as damage could occur to the fuel injector.

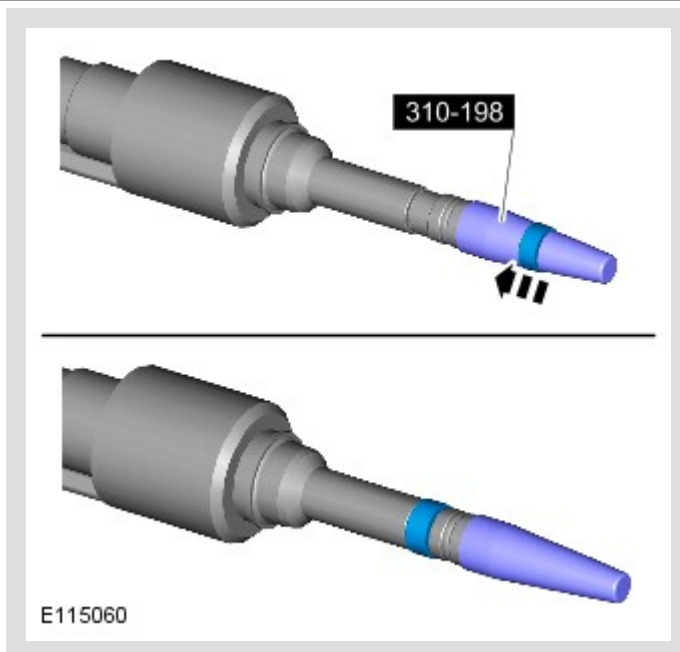


INSTALLATION

1.

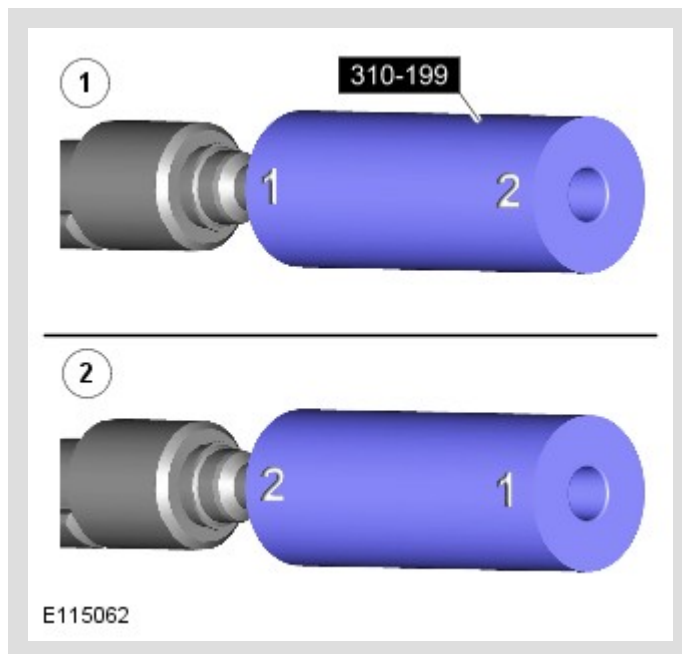


2.



Special Tool(s): [310-198](#)

3.

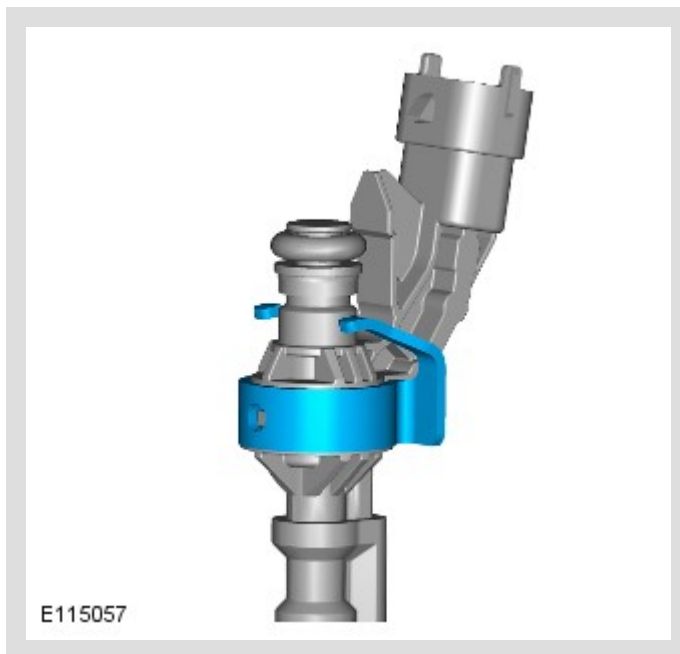


Special Tool(s): [310-199](#)

4.

⚠ CAUTION:

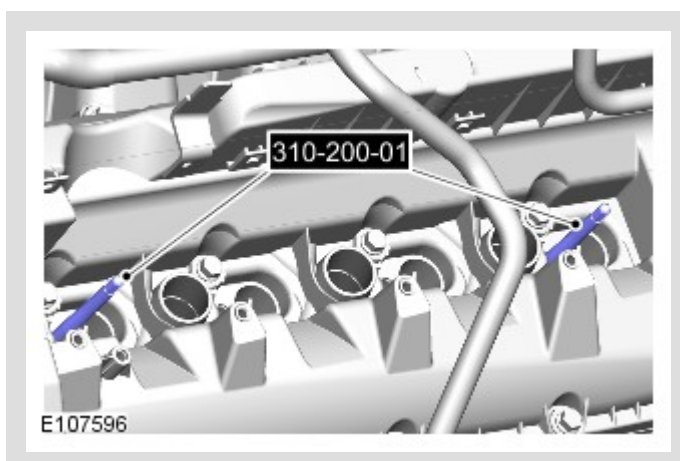
If the original fuel injector is being installed, the original fuel injector clamp must be installed with the fuel injector it was removed with.



5.

⚠ CAUTION:

If a new cylinder head has been installed then the special tool 310-200-02 without the thread must be used to install the fuel rail.

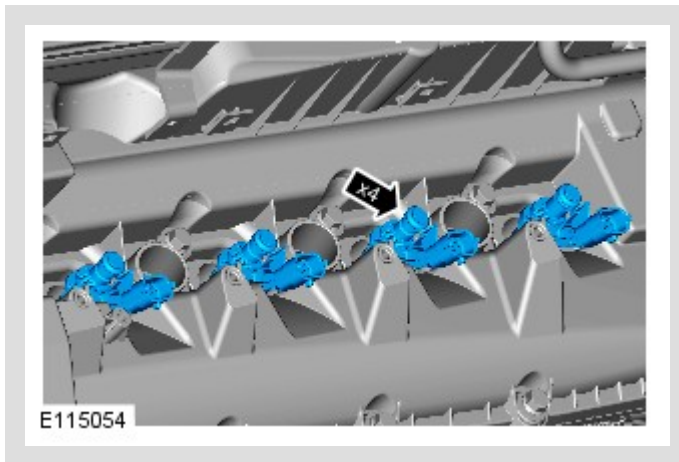


Special Tool(s): [310-200-01](#) , [310-200-02](#)

6.

⚠ CAUTIONS:

- Make sure that the area around the open fuel injector ports are clean and free of foreign material and lubricant prior to installing the fuel injector.
- When installing the fuel injector(s), make sure that the Teflon seal is clean and free of foreign material and lubricant.



7.



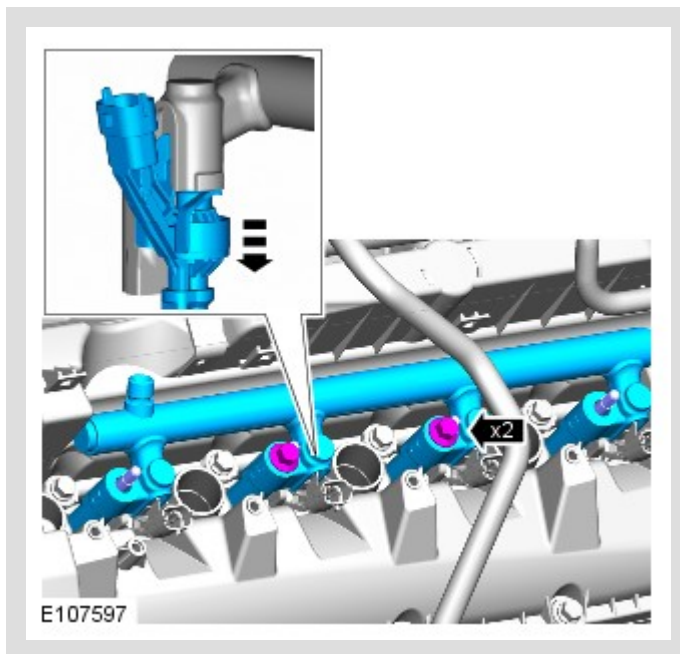
CAUTIONS:

- If new fuel injectors are installed, a new injector clamp must be installed.
- Make sure that the fuel injector is aligned and installed into the fuel rail correctly, as noted in the removal step.
- Tighten the fuel rail retaining bolts a turn at a time until the correct torque is achieved.



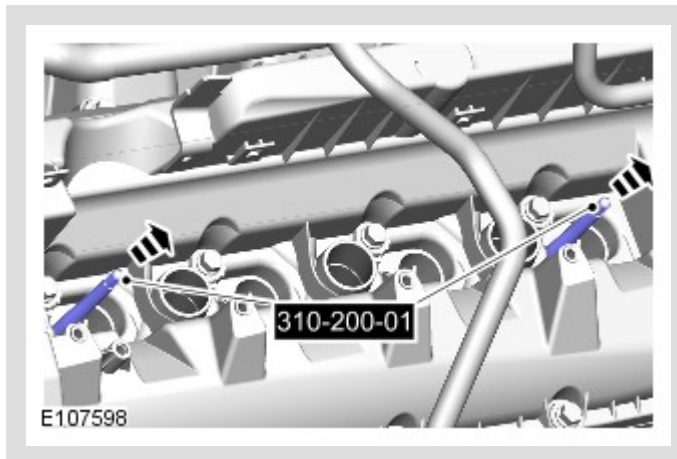
NOTE:

Lubricate the fuel injector O-ring seals with clean engine oil.



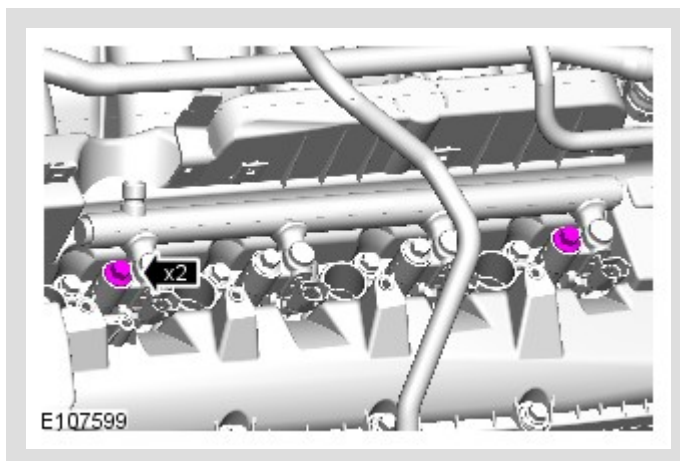
Torque: 20 Nm

8.



Special Tool(s): [310-200-01](#) , [310-200-02](#)

9.

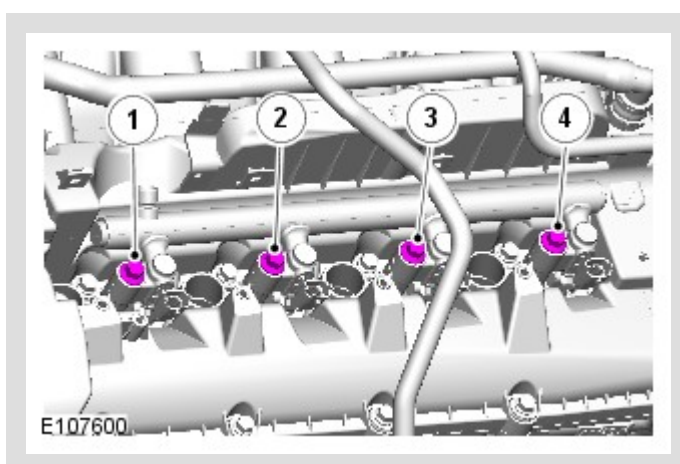


Torque: 20 Nm

10.

⚠ CAUTION:

Tighten the bolts in the sequence shown.



Torque:

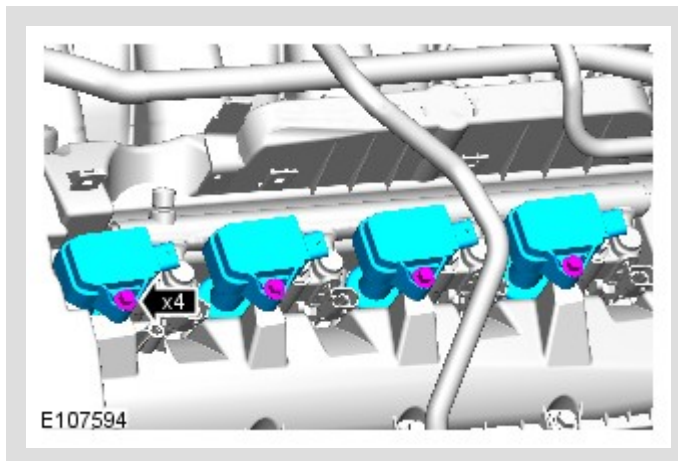
Bolt 2 30 Nm

Bolt 3 30 Nm

Bolt 1 30 Nm

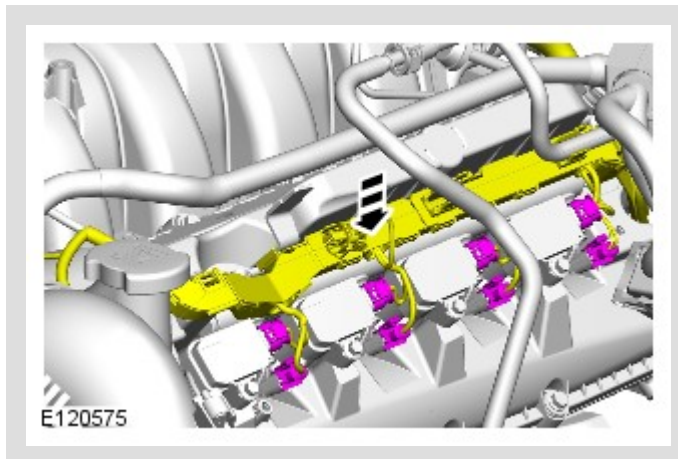
Bolt 4 30 Nm

11.



Torque: 8 Nm

12.



13.

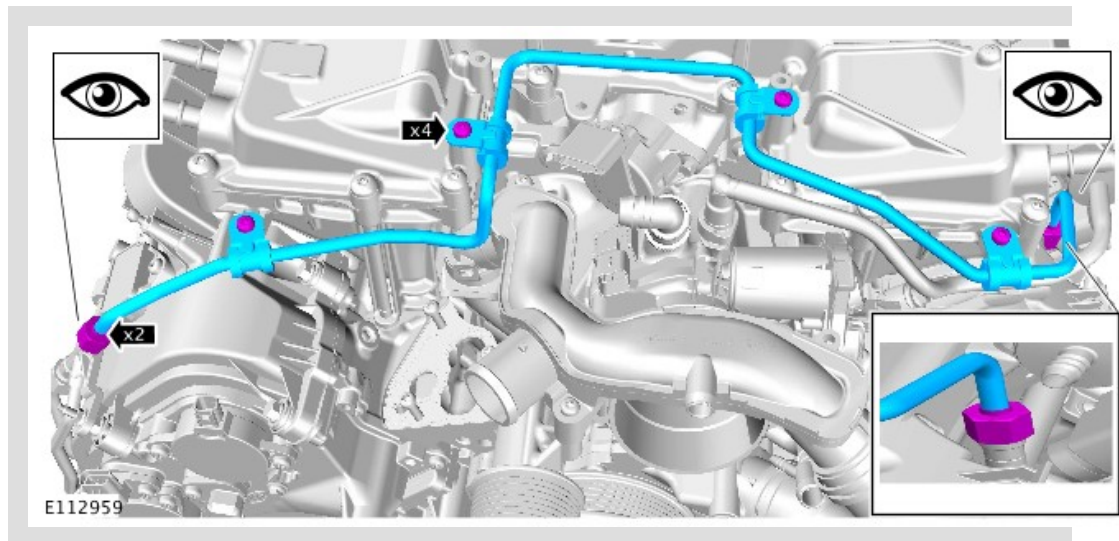
⚠ CAUTION:

Lubricate only the union threads with clean engine oil.

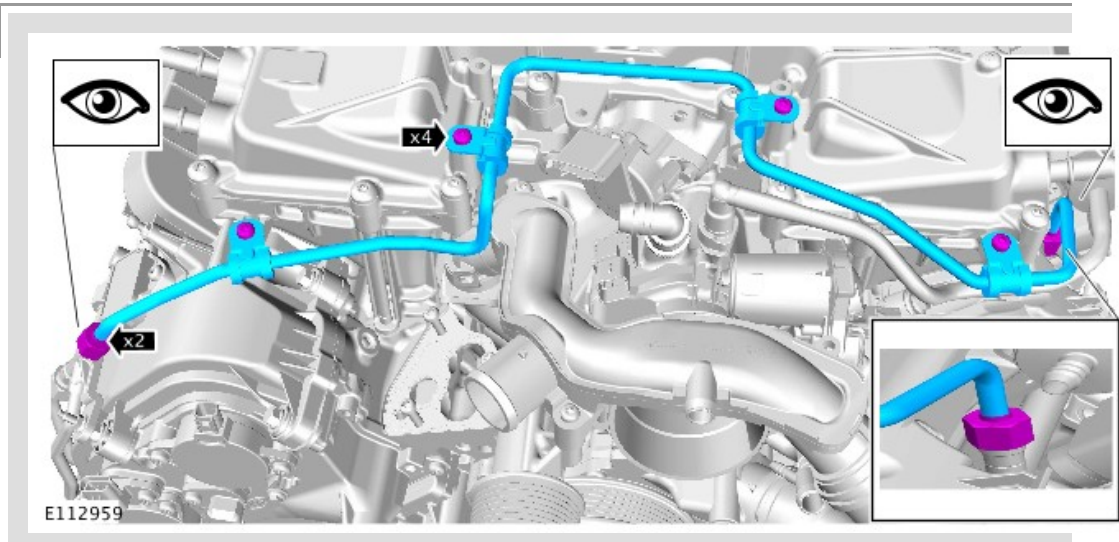
△ NOTES:

- Do not tighten at this stage.

- Remove and discard the blanking caps.



14.

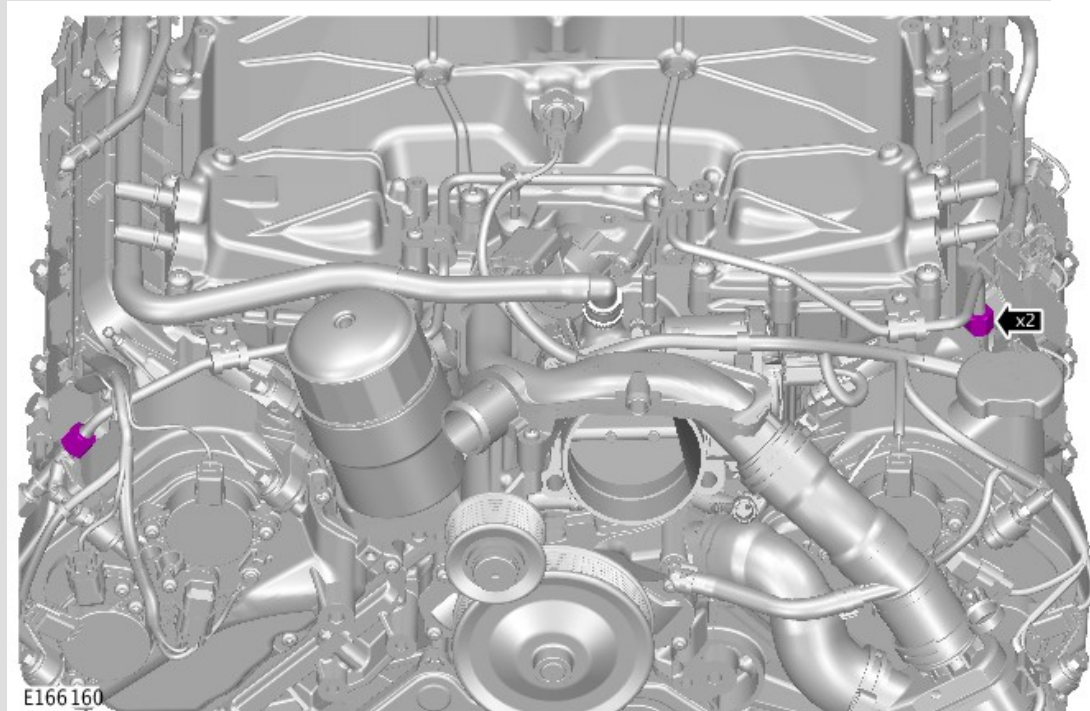


Torque:

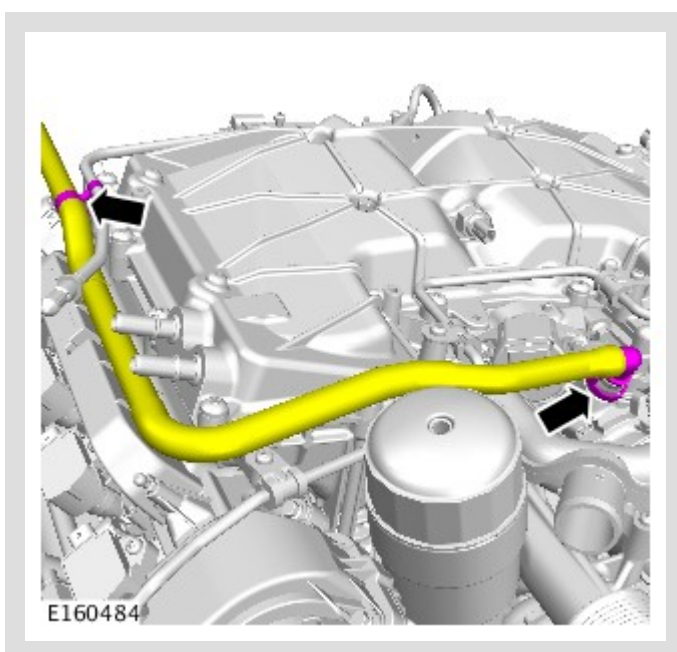
Unions **20 Nm**

Bolts **8 Nm**

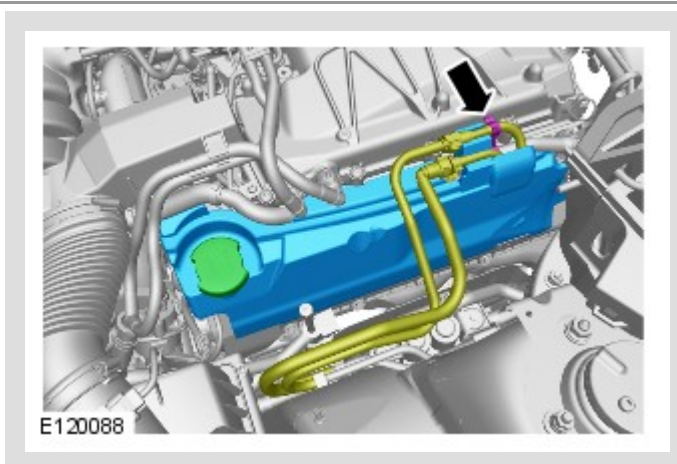
15.



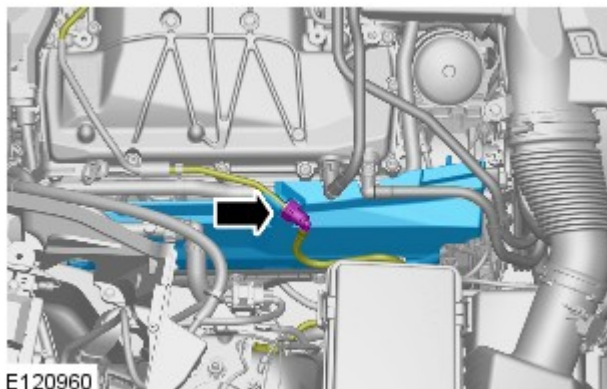
Torque: 21 Nm



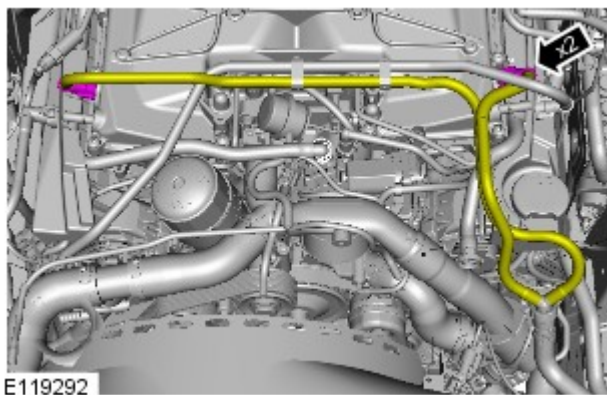
17.



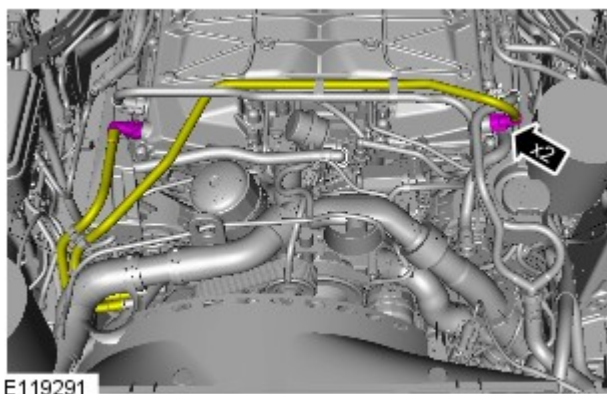
18.



19.



20.



21. Refer to: Air Cleaner Outlet Pipe T-Connector (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation).

22. Connect the battery ground cable.
Refer to: Specifications (414-00, Specifications).